

ThermoNAS - Problem

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ThermoNAS is used to solve a two-dimensional problem of heat transfer in a flat porous TPMS-channel (the porous medium is based on the topology of triply periodic minimal surfaces) during the flow of an incompressible liquid. Figure 1 shows the problem scheme.

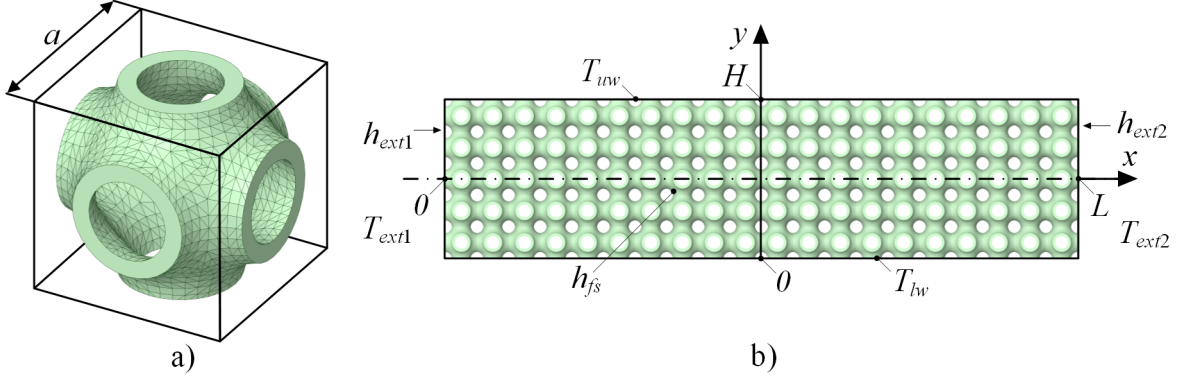


Figure 1: Scheme: a) Schwarz P TPMS unit cell; b) TPMS porous medium

Below is a list of assumptions regarding the mathematical formulation of the problem:

- Thermal dispersion is not taken into account due to low flow rates;
- No phase change;
- The y-component of velocity is not taken into account since laminar Darcian flow is considered;
- Fluid density and viscosity are constant;
- A steady flow of liquid is considered.

Taking into account the assumptions, the final system of equations, including initial and boundary conditions, will have the following form:

$$(1 - \phi)(\rho c_p)_s \frac{\partial T^s}{\partial t} = k_s \left[\frac{\partial^2 T^s}{\partial x^2} + \frac{\partial^2 T^s}{\partial y^2} \right] + A_0 h_{fs} (T^f - T^s); \quad (1)$$

$$\phi(\rho c_p)_f \left[\frac{\partial T^f}{\partial t} + u_x \frac{\partial T^f}{\partial x} \right] = k_f \left[\frac{\partial^2 T^f}{\partial x^2} + \frac{\partial^2 T^f}{\partial y^2} \right] + A_0 h_{fs} (T^s - T^f); \quad (2)$$

$$u_x = -\frac{K}{\mu} \frac{dP}{dx}; \quad (3)$$

$$\frac{du}{dx} = 0. \quad (4)$$

$$T^s(x, y, 0) = T^f(x, y, 0) = T_0; \quad (5)$$

$$-k_{ms} \frac{\partial T^s(x, y, t)}{\partial x} \Big|_{x=0} = h_{ext1} (T_{ext1} - T^s(0, y, t)); \quad (6)$$

$$k_{ms} \frac{\partial T^s(x, y, t)}{\partial x} \Big|_{x=L} = h_{ext2} (T_{ext2} - T^s(L, y, t)); \quad (7)$$

$$T^f(0, y, t) = T_{ext1}; \quad (8)$$

$$\left. \frac{\partial T^f(x, y, t)}{\partial x} \right|_{x=L} = 0; \quad (9)$$

$$T^f(x, 0, t) = T_{lw}; \quad (10)$$

$$T^f(x, H, t) = T_{uw}; \quad (11)$$

$$T^s(x, 0, t) = T_{lw}; \quad (12)$$

$$T^s(x, H, t) = T_{uw}; \quad (13)$$

$$P(0, y) = P_0; \quad (14)$$

$$u(0, y) = u_0. \quad (15)$$

where T^f , T^s are desired functions of temperature of fluid and solid; t is time; x , y are coordinates; ρ is density; c is specific heat; ϕ is porosity; k_s , k_f are effective thermal conductivities of solid and fluid; k_{ms} , k_{mf} are thermal conductivities of solid and fluid A_0 is effective heat transfer area; h_{fs} is fluid-solid heat transfer coefficient; u is velocity; K is permeability; P is pressure; μ is viscosity; h_{ext1} , h_{ext2} are heat transfer coefficients at the boundaries of a solid at points $x = 0$ and $x = L$; T_{ext1} , T_{ext2} are temperatures of the environments; T_0 is initial temperature; P_0 is initial pressure; u_0 is inlet velocity; T_{lw} , T_{uw} are constant temperatures, which are set at points $y = 0$ and $y = H$ for both liquid and solid; L is length of channel; H is height of channel.